RESEARCH STATEMENT

# Physically Grounded and Generalizable Robot Intelligence

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My research asks how robots can reuse experience when the physical conditions that made a behavior effective change. A familiar motion may have a different consequence on a more compliant surface, with a different body, or in a shared workspace. Generative models and reinforcement learning can supply useful behavioral alternatives; realizing those alternatives requires reasoning about the current robot and its environment. I want to build robot intelligence that combines these capabilities, making accumulated knowledge useful through local physical understanding and feedback.

My central hypothesis is that **physical structure provides an interface between reusable behavioral knowledge and realizable action**. My work on generative optimization investigates how dynamics, constraints, geometry, and contact feedback should shape behavior generation. Complementary work on federated reinforcement learning studies how distinct learners can exchange experience while retaining local specialization. These foundations motivate three connected directions: physical intelligence, adaptation across embodiments, and collaborative autonomy (Figure 1).

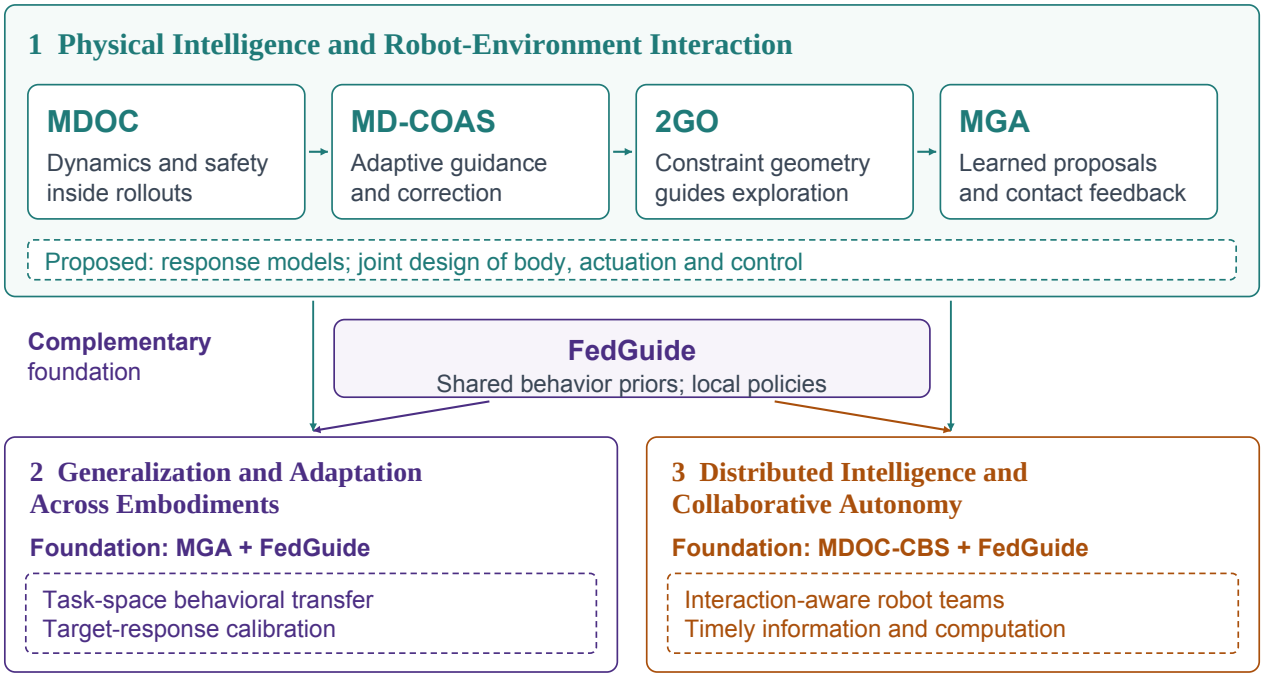



**Figure 1.** Research roadmap. The four-paper progression makes dynamics, adaptive feasibility, geometry, and contact feedback increasingly central to behavior generation. FedGuide provides a complementary foundation for sharing behavioral knowledge. Solid regions identify existing foundations; dashed regions identify proposed research.

## 1 Physical Intelligence and Robot–Environment Interaction

Physical intelligence requires a robot to reason about how actions become outcomes. Many tasks admit several plausible motions, yet dynamics and environmental constraints make only some of them useful. **I study how physical constraints can inform generation from its beginning.**

In *MDOC* [1], dynamics and barrier-based corrections enter candidate rollouts before their outcomes guide refinement. This brings feasibility into the information used to construct behavior and supports multi-robot planning through local physical optimization coupled to conflict resolution. Enforcing feasibility also creates a search problem: correcting every candidate aggressively can suppress alternatives that are temporarily infeasible but informative. In *MD-COAS* [2], we therefore combine soft feasibility guidance with local correction and adapt their strength and computational effort during search. Together, these works make constraint handling an active part of generative reasoning (Figure 2).

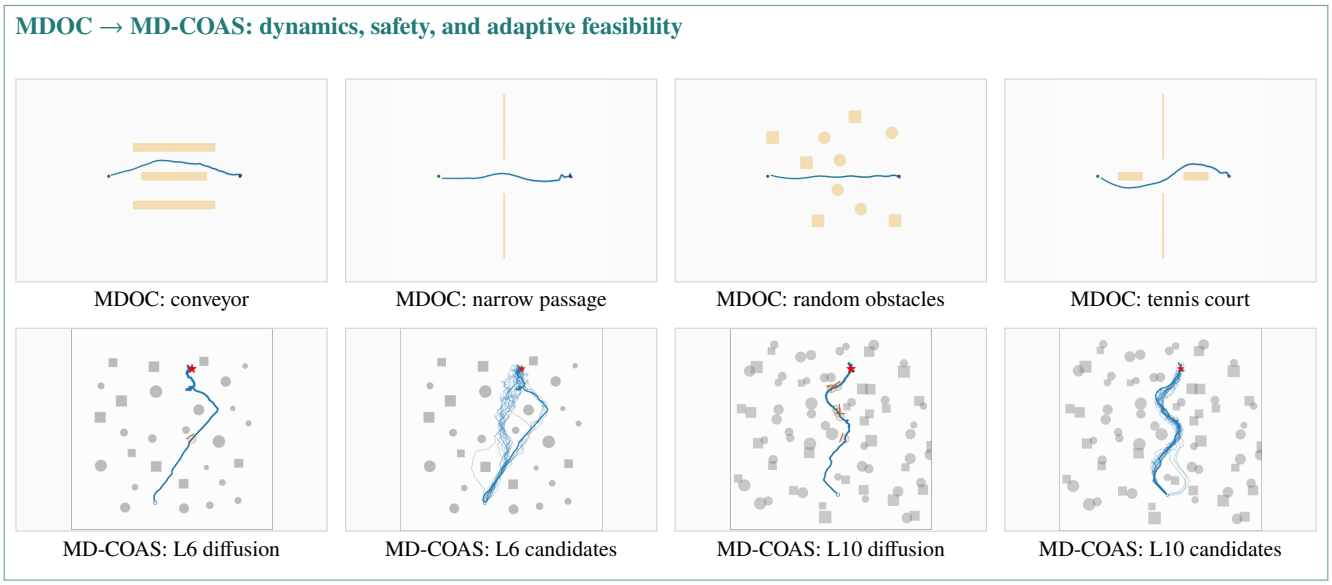



**Figure 2.** Direction 1: the complete MDOC single-robot repertoire and both MD-COAS two-dimensional planning settings, pairing generated paths with candidate alternatives. The arm planning and execution views continue in Figure 3.

Adaptive enforcement determines when and how strongly feasibility should constrain exploration; the geometry of the feasible region also determines which changes are useful. In *2GO* [3], we use active constraint geometry to shape refinement directions and stochastic exploration. Disconnected footholds and coupled body clearance illustrate why discovering useful alternatives depends on the organization of the search itself. Contact makes this relationship deeper: the same displacement command can produce motion, deformation, or force, depending on impedance and the environment. *MGA* [4] brings learned behavioral proposals, model-based reassessment, and feedback-conditioned geometric refinement into a common motion–impedance decision space. Learned experience supplies task-informed alternatives, while physical models and current feedback determine their relevance. During surface interaction, for example, the robot must revise how it moves as contact changes the relationship between commanded displacement and realized motion. This advances the progression from admissible trajectories toward behavior that accounts for its physical consequences (Figures 3–5).

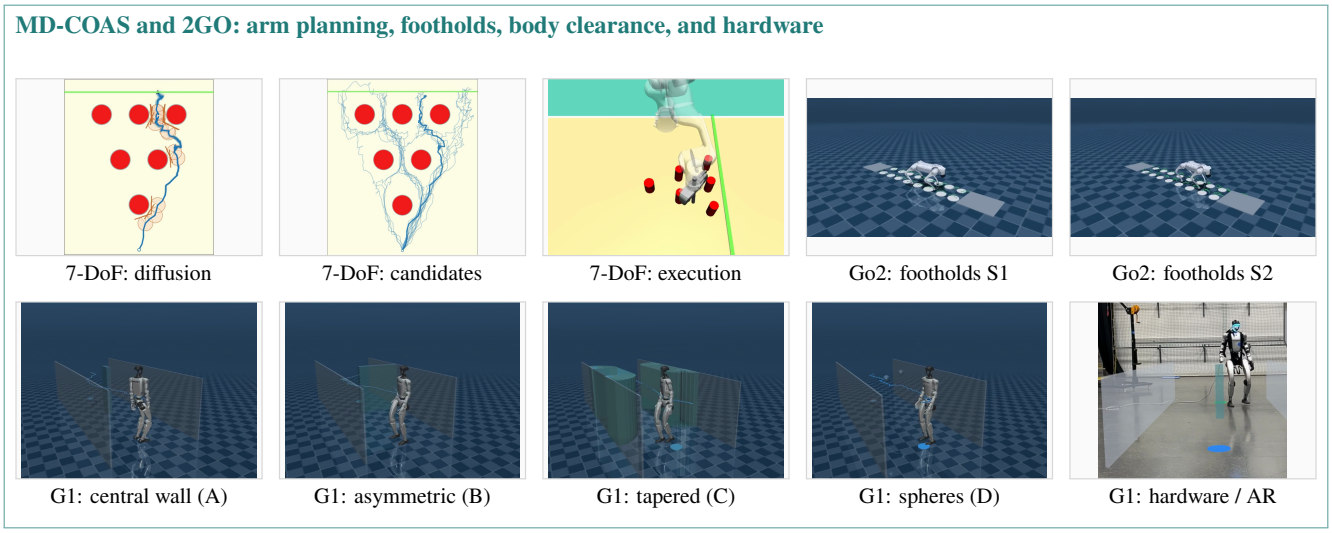



**Figure 3.** Direction 1: MD-COAS arm planning and the complete 2GO repertoire. The top row shows 7-DoF diffusion, candidate paths, an independent execution run, and both stepping-stone tasks. The bottom row shows all four simulated G1 corridor variants and hardware with registered AR obstacles.

Together, these works develop a view of behavior generation in which **models predict consequences, adaptive constraints regulate feasibility, geometry organizes alternatives, and learned proposals are reassessed through interaction**. The remaining challenge is to determine which aspects of physical response must be represented for a robot to revise its behavior under unfamiliar contact conditions. The surface interactions in Figure 4 make these variations concrete.

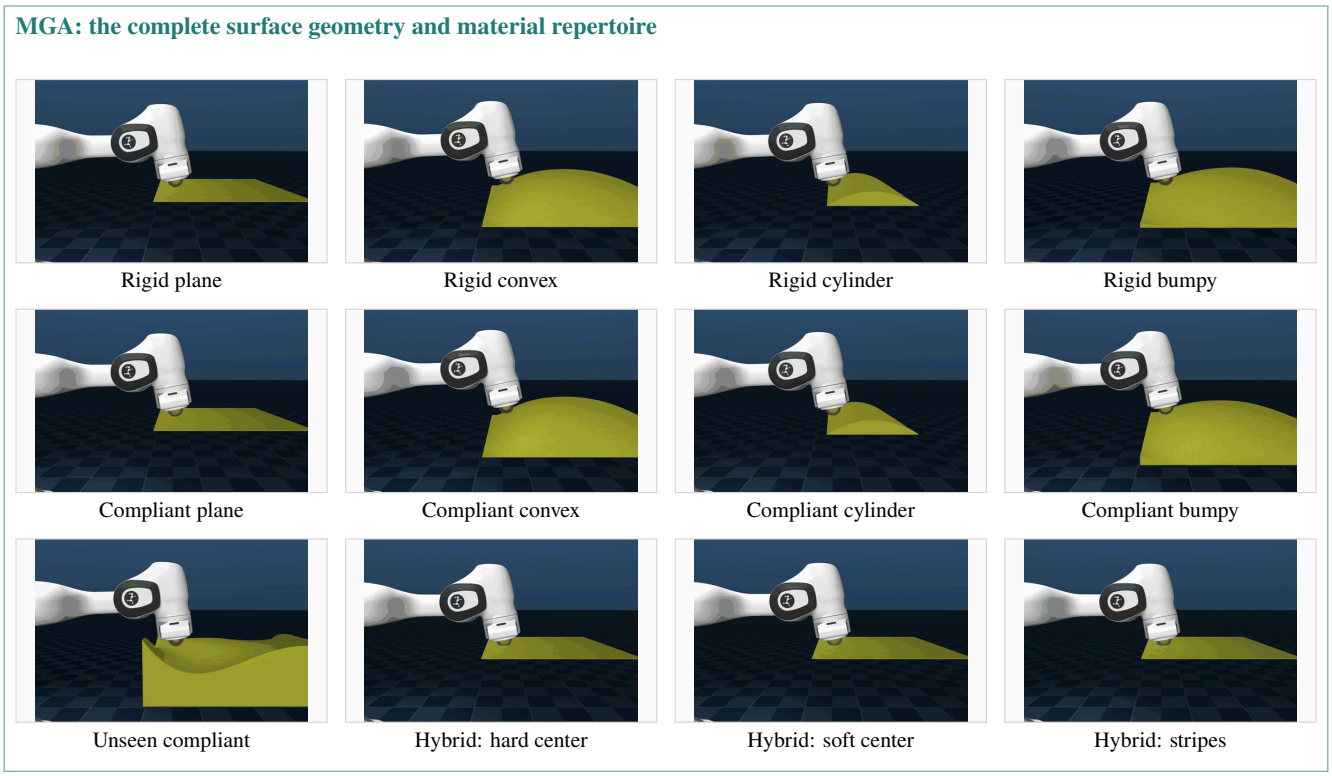



**Figure 4.** Direction 1: the complete MGA surface repertoire, spanning rigid, compliant, unseen, and hybrid material conditions. Rigid arms interact with varying material compliance, motivating the response-modeling and co-design questions in the text.

I want to address this challenge with compact response models that learn how changing compliance, actuation, and contact transitions affect task progress on a given robot. Coupled to behavior priors, such models could guide whether to retain a familiar strategy, alter its impedance, or explore a different contact mode.

Feedback would update both the response model and the confidence placed in its refinement. This would make interaction a source of information for behavior generation and recovery, connecting robot learning with system identification and feedback control. Figure 5 extends this interaction perspective across manipulation, whole-body contact, and hardware.

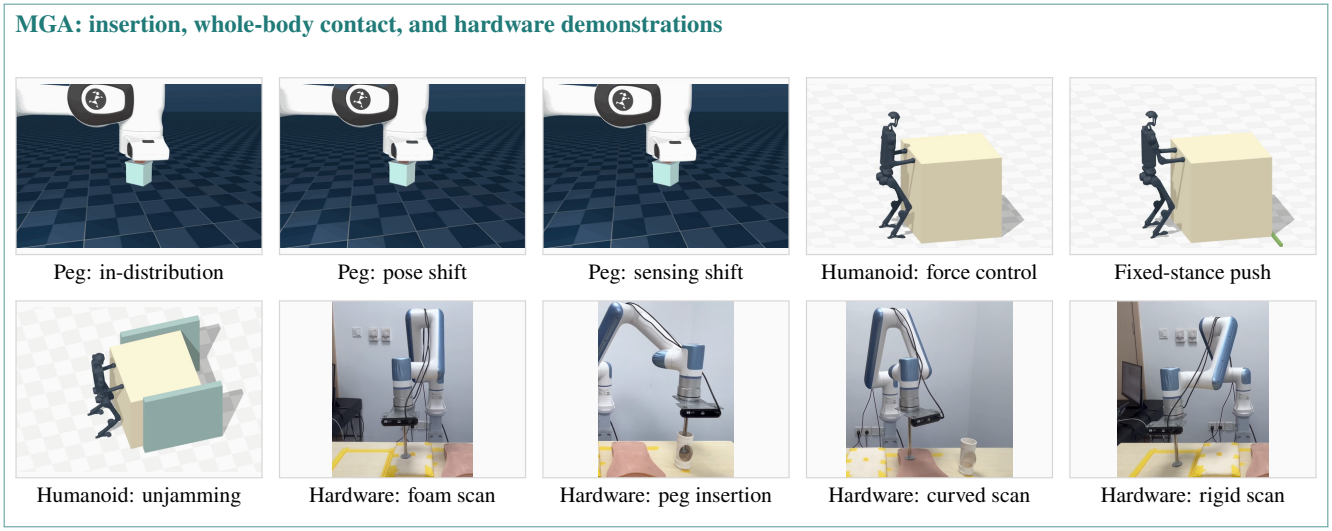



**Figure 5.** Direction 1: all remaining MGA task and hardware demonstrations in the original gallery. The insertion and humanoid panels are simulation; the final four panels are hardware. Hardware uses model-based refinement and force feedback.

The same reasoning extends to the physical capabilities available to a controller. If morphology and material response determine which behaviors can be realized, they should participate in the search. I want to investigate joint generation of body, actuation, and control, beginning with soft-robot co-design.

A key difficulty is deciding when an inexpensive model preserves useful design comparisons and when more detailed physics could change the choice. Combining learned proposals with physical evaluation at multiple fidelities offers a path toward exploring this coupled design space. My longer-term aim is to understand how learning, feedback, and embodiment can jointly produce capabilities that are difficult to obtain by optimizing each in isolation.

## 2 Generalization and Adaptation Across Embodiments

A behavior that is physically realizable for one robot may fail on another because the same command produces a different response. Changes in actuation, compliance, sensing, and contact can alter how a robot accomplishes an otherwise unchanged task. This realization gap raises a question that extends beyond learning a policy for each platform: **what behavioral knowledge should remain reusable, and what must adapt to the target body?** I view generalization as preserving useful task structure while revising how it is physically realized. The challenge is to retain alternative ways of accomplishing a task without assuming that their motions or control parameters transfer unchanged.

My work on *FedGuide* [5], with Professor Gauri Joshi, provides a foundation for this separation. Heterogeneous learners can develop complementary behaviors, yet combining their policy parameters can obscure the alternatives that make each learner effective. FedGuide shares diffusion behavior priors while retaining local policies, using optimal-transport expert routing to combine behavioral knowledge and value-baseline guidance to support local learning. Its contribution to this agenda is a way to communicate possibilities while allowing each learner to determine how to act.

The work studies heterogeneous dynamics and rewards within shared state and action spaces; extending this principle across embodiments requires an interface that relates shared task intent to different physical responses (Figures 6–8).

## 2 Shared behavioral knowledge with local specialization

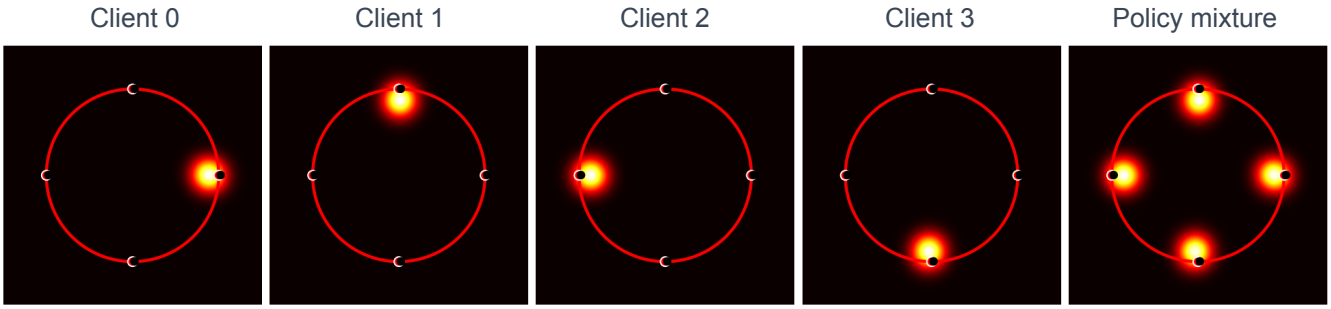



**Figure 6.** Direction 2: original FedGuide policy densities show four distinct local behaviors and their mixture. Sharing behavioral knowledge while preserving specialization motivates the cross-embodiment extension below.

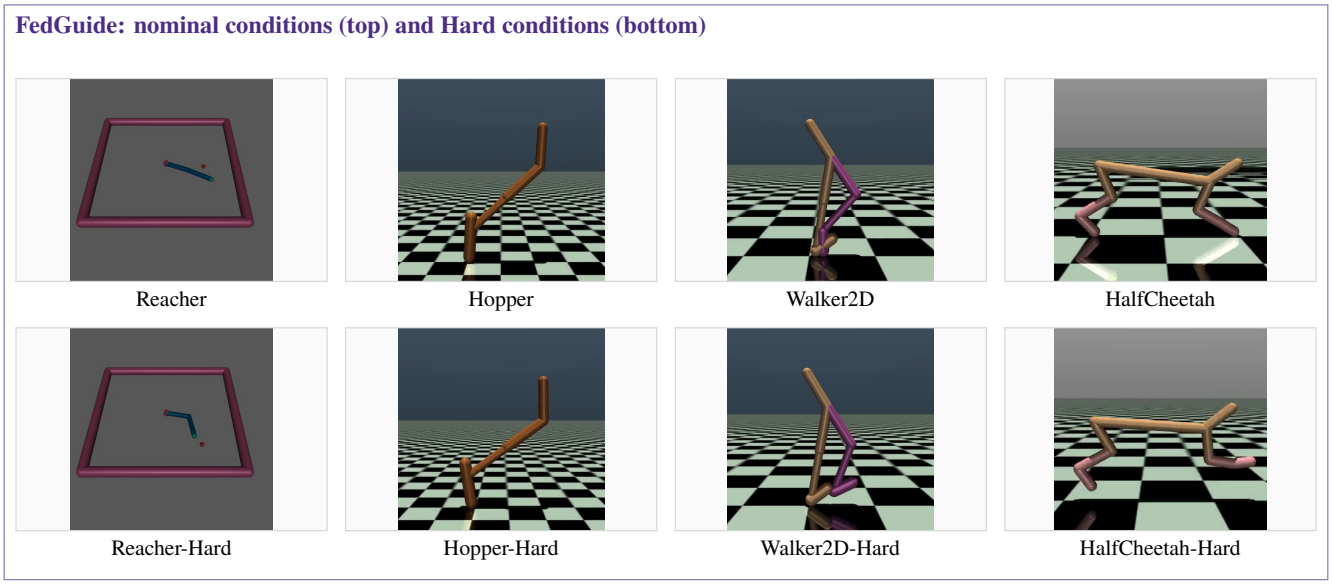



**Figure 7.** Direction 2: representative FedGuide policy rollouts for every reaching and locomotion task, including each Hard setting. Clients vary in local dynamics and rewards; each panel shows one selected client execution.

I want to develop this interface through distributions over task progress, end-effector motion, and contact intent, coupled to compact models of the target body's response. For example, an insertion strategy may remain useful across arms while its timing, displacement, and impedance require adjustment. Limited target interactions could identify the response differences relevant to that adjustment, allowing sampled behaviors to be refined through local models and feedback. Interaction should be purposeful: a probe is valuable when it distinguishes physical explanations that would lead to different actions. The complete shared-embodiment manipulation repertoire is shown in Figure 8.

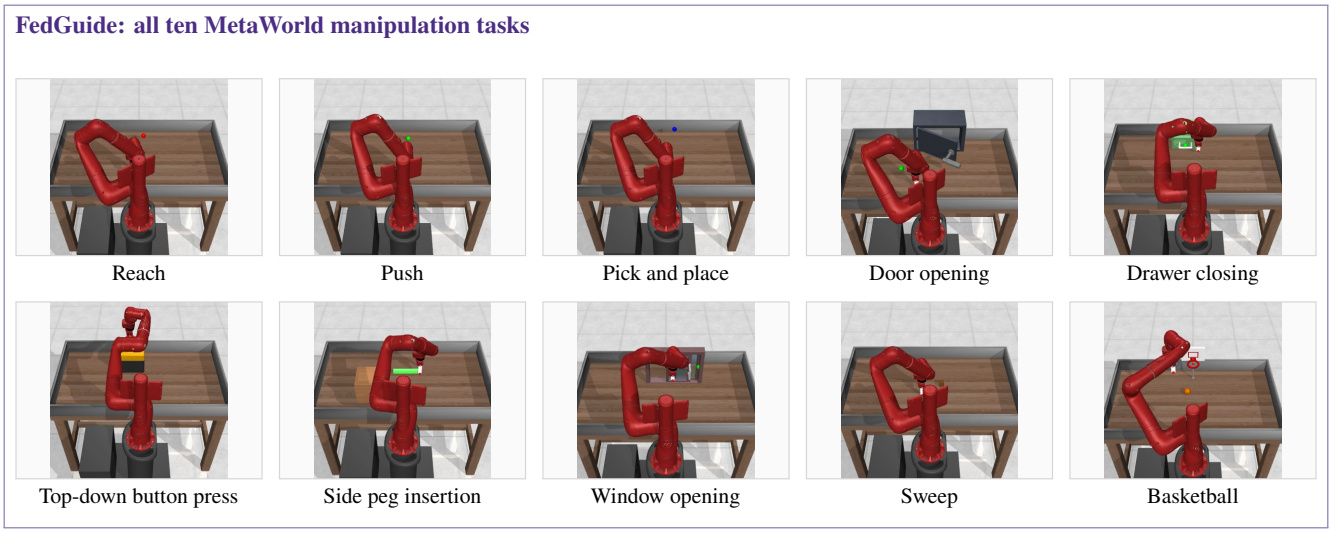



**Figure 8.** Direction 2: representative policy executions for all ten MetaWorld tasks in the FedGuide gallery. Each panel shows a selected local-policy rollout on the shared robot embodiment. Transfer across embodiments is proposed.

The underlying scientific question is which aspects of behavior are invariant across embodiments and which must remain adaptable. By studying controlled changes in physical response and then transfer between robot platforms, I hope to establish when shared behavioral structure reduces the need for new experience, and when unfamiliar physical consequences demand additional learning.

## 3 Distributed Intelligence and Collaborative Autonomy

Individually executable behaviors do not automatically compose into a team's ability to accomplish a task. Robots can interfere through shared space, contact, timing, and limited resources, even when each follows a valid local plan. My work provides two complementary foundations for addressing this dependence. MDOC-CBS [1] connects local dynamics-aware planning with shared conflict resolution, while FedGuide [5] enables heterogeneous learners to exchange behavioral knowledge while retaining distinct local policies. These perspectives motivate **collaborative autonomy that preserves local physical responsiveness while reasoning about joint consequences**. Extending them to cooperation requires understanding how one robot's action changes what another can do (Figure 9).

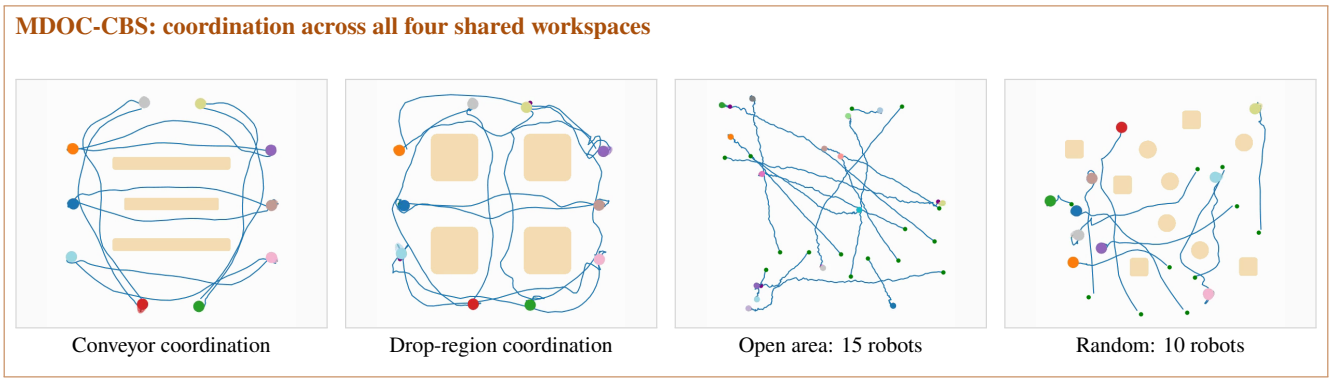



**Figure 9.** Direction 3: all four MDOC-CBS coordination scenes, combining dynamics-aware local motion and shared conflict resolution. These provide the foundation for the interaction-aware teamwork proposed below.

I want to study cooperation in which robots stabilize an object for one another or prepare a shared workspace for a subsequent action. I propose joint execution models that describe how concurrent skills affect contact, available resources, and task progress.

A central question is how much information about a robot's body, controller, and execution state must be retained to predict whether skills compose reliably. Calibration would become an informative action within team decision-making: a short interaction could resolve uncertainty that would change the choice or timing of a shared plan. When a teammate or controller changes, these models could help identify which interactions require new evidence and which existing knowledge remains useful.

Communication and computation also determine whether shared knowledge can improve a physical decision. A prediction may arrive after the relevant contact transition, and a useful behavior prior may require more computation than a robot can afford before acting. My ongoing EdgeCloud work motivates this systems perspective on autonomy. I want to study when robots should exchange predictions, request shared computation, or rely on local feedback, with these choices informed by the decisions they can still affect. This connects distributed learning and computing with the timing of physical interaction, and makes information availability part of how a team reasons about action.

Through GenerativeDynamics, I am building common interfaces across tasks, solvers, and execution to support this agenda. My doctoral goal is to develop principles connecting generative learning, physical models, and autonomous systems: robots should accumulate reusable knowledge, adapt its realization through interaction, and improve collectively while retaining their complementary capabilities.

**References**

[1] Z. He, Y. Shaoul, and J. Li. Model-Based Diffusion Optimal Control for Multi-Robot Motion Planning. 2026.

[2] Z. He et al. Motion Planning with Model-Based Diffusion via Constraint Optimization and Adaptive Scheduling. 2026.

[3] Z. He, X. Wang, and C. Liu. Model-Based Geometry-Aware Generative Optimization for Constrained Locomotion Planning (2GO). 2026.

[4] Manifold Generative Annealing for Contact Constraint-aware Impedance Control with Reinforcement Learning Prior (MGA). Research manuscript.

[5] Z. He and G. Joshi. FedGuide: Diffusion-Prior Alignment and Value Baseline Guidance for Heterogeneous Federated Reinforcement Learning. 2026.